(Please write your Enrollment No. immediately)

Enrollment No: ……

SUPPLEMENTARY MID-TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Second Semester, March 2025

Paper Code: BS106 **Subject: Applied Physics-II**

Time: 1½Hrs. **Maximum Marks: 30**

Note: Attempt Question no 1, which is compulsory, and any two more questions from the remaining

Q. No.	Question 1	Marks	CO
1(a)	Calculate the De-Broglie wavelength of 40KeV electron	[2.5]	CO1
1(b)	Find the average speed and root mean square speed for an assembly of two molecules, one with a speed of 1m/s and the other with a speed of 3m/s.	[2.5]	CO2
1(c)	Give physical interpretation of wave function ψ and its normalization.	[2.5]	CO1
1(d)	The Fermi level in potassium is 2.1 eV at a particular temperature. Calculate the number of free electrons per unit volume in potassium at the same temperature.	[2.5]	CO2
	Question 2		
2(a)	Set up the Schrodinger wave equation for a particle in an infinite well (1D). Obtain the wave functions and energy eigenvalues.	[5]	CO1
2(b)	What is Quantum Mechanical Tunneling	[2]	CO1
2(c)	An electron has a speed 500m/s accurate to 0.01%. With what fundamental accuracy can we locate the position of an electron?	[3]	CO1
	Question 3		
3(a)	What is Black Body Radiation? Derive Planck's formula for the energy distribution in the spectrum of a Black body.	[6]	CO2
3(b)	Fermi Energy in Ag is 5.51eV. What is the average energy of free electrons in Ag at 0K? What temperature is necessary for the average molecular energy in an ideal gas to have this value?	[4]	CO2
	Question 4		
4(a)	Compare the salient features of Maxwell Boltzmann(MB), Bose Einstein(BE) and Fermi Dirac(FD) Statistics	[5]	CO2
4(b)	Calculate the expectation value of the momentum of a particle trapped inside a 1D box.	[2]	CO1
4(c)	What is zero point energy in quantum mechanics? The energy of the linear Harmonic Oscillator in the third excited state is 0.1eV. Find the frequency of vibration.	[3]	CO1